

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q22-2124

Redman 138 kV
January 21, 2024

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2124. The Interconnection Customer proposes a 195.1 MW natural gas combined cycle facility interconnecting at the Redman 138 kV substation in the West Virginia area. The request was received on June 24, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 582 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 23 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$26,156,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Arvin - Garden Corners 138.0 kV
Contingency: Douglas City - Southwest Research Institute 138.0 kV
Loading: 126.8% (Rating: 247 MVA)
DFAX: 18.8%
Pre-project loading: 112.0%
2. Facility: Millville - Sleepy Hollow 138.0 kV line
Contingency: Clark County - Kearneyville 345.0 kV
Loading: 120.6% (Rating: 616 MVA)
DFAX: 24.9%
Pre-project loading: 112.7%
3. Facility: Douglas City - Millville 138.0 kV
Contingency: Clark County - Kearneyville 345.0 kV
Loading: 119.0% (Rating: 613 MVA)
DFAX: 24.9%
Pre-project loading: 111.0%
4. Facility: Carlota - Arvin 138.0 kV
Contingency: Arvin - Shuffleton 138.0 kV line
Loading: 117.8% (Rating: 198 MVA)
DFAX: 16.4%
Pre-project loading: 101.6%
5. Facility: Claytonia - Shuffleton 138.0 kV
Contingency: Clark County - Kearneyville 345.0 kV
Loading: 114.5% (Rating: 329 MVA)
DFAX: 20.9%
Pre-project loading: 102.1%

6. Facility: Douglas City Substation - Garden Corners 138.0 kV
Contingency: Douglas City - Southwest Research Institute 138.0 kV
Loading: 112.7% (Rating: 322 MVA)
DFAX: 18.8%
Pre-project loading: 101.4%
7. Facility: Arvin - Shuffleton Substation 138.0 kV
Contingency: Clark County Substation - Kearneyville 345.0 kV
Loading: 112.2% (Rating: 370 MVA)
DFAX: 20.9%
Pre-project loading: 101.1%
8. Facility: Air Force Base - Tiger Bay 345.0 kV
Contingency: Air Force Base 345/138kV Transformer
Loading: 110.6% (Rating: 1,124 MVA)
DFAX: 12.0%
Pre-project loading: 108.5%
9. Facility: Lcec No. 5 - Tanner 138.0 kV
Contingency: Glenbrook - Redman 138.0 kV Line
Loading: 110.3% (Rating: 386 MVA)
DFAX: 7.4%
Pre-project loading: 106.5%
10. Facility: Arvin - Southwest Research Institute 138.0 kV
Contingency: Kearneyville - Clark County 345.0 kV
Loading: 108.1% (Rating: 341 MVA)
DFAX: 15.1%
Pre-project loading: 99.5%
11. Facility: Palmyra - Grimmway-Malaga 138.0 kV
Contingency: Clark County - Kearneyville 345.0 kV
Loading: 107.8% (Rating: 565 MVA)
DFAX: 47.2%
Pre-project loading: 91.4%
12. Facility: Glenbrook - Palmyra 138.0 kV
Contingency: Kearneyville - Clark County 345.0 kV
Loading: 107.3% (Rating: 580 MVA)
DFAX: 47.2%
Pre-project loading: 91.4%
13. Facility: Redman - Glenbrook Substation 138.0 kV L
Contingency: Clark County - Kearneyville 345.0 kV L
Loading: 106.9% (Rating: 594 MVA)
DFAX: 47.2%
Pre-project loading: 91.4%
14. Facility: Douglas City - Southwest Research Institute 138.0 kV
Contingency: Clark County - Kearneyville 345.0 kV
Loading: 106.5% (Rating: 385 MVA)
DFAX: 15.1%
Pre-project loading: 98.8%
15. Facility: Claytonia - Scurry 138.0 kV
Contingency: Loudon - Thomaston 345.0 kV Line
Loading: 106.4% (Rating: 198 MVA)
DFAX: 7.6%
Pre-project loading: 99.0%
16. Facility: Artesia Country Club - Ross Fork Substation 138.0 kV line
Contingency: Air Force Base - Artesia Country Club 345.0 kV line

Loading: 106.0% (Rating: 1,009 MVA)

DFAX: 9.0%

Pre-project loading: 104.3%

17. Facility: Sleepy Hollow - Grimmway-Malaga 138.0 kV Line

Contingency: Karneyville - Clark County 345.0 kV

Loading: 104.6% (Rating: 725 MVA)

DFAX: 24.9%

Pre-project loading: 97.9%

18. Facility: Lamont - Frderick 138.0 kV

Contingency: Low Gap - Jack Mcdonough 138.0 kV line

Loading: 104.6% (Rating: 474 MVA)

DFAX: 5.3%

Pre-project loading: 102.4%

19. Facility: Tiger Bay 345/138kV Autotransformer

Contingency: Tiger Bay - Dranesville 345.0 kV

Loading: 104.1% (Rating: 549 MVA)

DFAX: 6.1%

Pre-project loading: 102.0%

20. Facility: G-P Gypsum - Brighton Substation 138.0 kV

Contingency: Artesia Country Club - Air Force Base 345.0 kV

Loading: 102.5% (Rating: 676 MVA)

DFAX: 5.3%

Pre-project loading: 101.0%

21. Facility: Columbus City 345/1kV XFMR

Contingency: International Paper Valliant Ok - Kearneyville 345.0 kV L

Loading: 102.0% (Rating: 755 MVA)

DFAX: 26.4%

Pre-project loading: 95.2%

22. Facility: Columbus City 138-1 kV TX

Contingency: International Paper Valliant Ok - Keareyville 345.0 kV

Loading: 101.7% (Rating: 772 MVA)

DFAX: 26.4%

Pre-project loading: 95.0%

23. Facility: Tanner - Gene 138.0 kV Line

Contingency: Redman - Glenbrook 138.0 kV

Loading: 100.2% (Rating: 416 MVA)

DFAX: 7.4%

Pre-project loading: 96.7%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Arvin - Garden Corners 138.0 kV	Rebuild line with higher-capacity conductor	\$10,007,000	46% (\$4,583,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Sleepy Hollow - Millville 138.0 kV	Rebuild line with higher-capacity conductor	\$5,824,000	22% (\$1,270,000)
Millville - Douglas City 138.0 kV line	Rebuild line with higher-capacity conductor	\$14,408,000	22% (\$3,143,000)
Carlota - Arvin 138.0 kV	Rebuild line with higher-capacity conductor	\$3,618,000	48% (\$1,720,000)
Claytonia - Shuffleton 138.0 kV	Rebuild line with higher-capacity conductor	\$4,627,000	35% (\$1,620,000)
Douglas City Substation - Garden Corners 138.0 kV	Rebuild line with higher-capacity conductor	\$1,543,000	46% (\$707,000)
Arvin - Shuffleton 138.0 kV	Rebuild line with higher-capacity conductor	\$2,796,000	35% (\$979,000)
Air Force Base - Tiger Bay 345.0 kV	Rebuild line with higher-capacity conductor	\$13,650,000	2% (\$252,000)
Tanner - Lcec No 5 138.0 kV L	Rebuild line with higher-capacity conductor	\$4,460,000	28% (\$1,242,000)
Arvin - Southwest Research Institute 138.0 kV	Reconductor line	\$2,486,000	56% (\$1,394,000)
Palmyra - Grimmway-Malaga 138.0 kV Line	Reconductor line	\$1,232,000	40% (\$492,000)
Glenbrook - Palmyra 138.0 kV Line	Reconductor line	\$962,000	40% (\$384,000)
Redman - Glenbrook Substation 138.0 kV L	Reconductor line	\$539,000	40% (\$215,000)
Douglas City - Southwest Research Institute 138.0 kV line	Reconductor line	\$799,000	56% (\$448,000)
Claytonia - Scurry 138.0 kV	Reconductor line	\$2,442,000	19% (\$465,000)
Artesia Country Club - Ross Fork Substation 138.0 kV line	Reconductor line	\$1,583,000	6% (\$88,000)
Sleepy Hollow - Grimmway-Malaga 138.0 kV Line	Reconductor line	\$956,000	22% (\$209,000)
Lamont - Frederick 138.0 kV	Reconductor line	\$3,238,000	6% (\$193,000)
Tiger Bay 345/138kV Autotransformer	Replace transformer with higher-rated unit	\$16,571,000	8% (\$1,407,000)
G-P Gypsum - Brighton Substation 138.0 kV	Reconductor line	\$2,050,000	6% (\$133,000)
Columbus City 345/1kV Autotransformer	Replace transformer with higher-rated unit	\$20,043,000	20% (\$4,058,000)
Columbus City 138-1 kV TX	Replace transformer with higher-rated unit	\$5,409,000	20% (\$1,095,000)
Tanner - Gene 138.0 kV Line	Reconductor line	\$212,000	28% (\$59,000)
	Total allocated to project		\$26,156,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.